

Identifying optimal measurement protocols for giving fertilization advice

Supervisors (TU Delft): Neil Budko, Corné Verburg

Supervisors (CropConsult): Jerke Eisma

Introduction

High-throughput phenotyping methods make it possible to collect an increasing amount of information about potato growth and environmental conditions. However, collecting more data is only useful if this information contributes to better management decisions. In practice, agronomists combine measurements, model outputs, weather information and expert knowledge to formulate fertilization advice. This decision process is often not explicitly documented, making it difficult to determine which measurements actually influence the advice and which provide little additional information. The goal of this project is to use historical data to make this advisory process more transparent, by identifying which measurements are most informative for reproducing fertilization advice and whether a smaller set of measurements could provide essentially the same information.

CropConsult, a company that provides advice to farmers, has collected field data containing soil measurements, crop biomass and nitrogen status, weather information, previous fertilizer applications, and the corresponding fertilization advice. These data provide an opportunity to investigate how strongly the recorded measurements are related to the advice that was eventually given, and whether there are measurements that provide little additional information once the other measurements are known.

Let A denote a fertilization advice and let $\mathbf{x} = (x_1, \dots, x_n)$ denote the information available at the time the advice was formulated. The advisory process can then be represented by the conditional distribution

$$p(A \mid \mathbf{x}). \quad (1)$$

The first goal is to determine how well the historical advice can be reproduced from the recorded information. Subsequently, the relevance of individual measurements can be studied by quantifying how much additional information a variable x_j provides about the advice once the remaining measurements are known. This can be done, for example, by comparing the conditional distributions

$$p(A \mid \mathbf{x}), \quad \text{and} \quad p(A \mid \mathbf{x}_{-j}), \quad (2)$$

which represent the advice given with and without the measurement x_j , respectively. The project will investigate how to quantify the difference between these two distributions and how to use this information to identify which measurements are most informative for reproducing the advice.

Reference for colloquium

J. R. Vergara and P. A. Estévez, “A review of feature selection methods based on mutual information,” *Neural Computing and Applications*, 24, 175–186, 2014.

Other references

I. Guyon and A. Elisseeff, “An introduction to variable and feature selection,” *Journal of Machine Learning Research*, 3, 1157–1182, 2003.